Adaptive Differential Privacy for Federated IoHT Intrusion Detection: A Privacy–Utility Study under Heterogeneous Clients

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***Abstract*—Federated learning (FL) is a promising approach for intrusion detection in Internet-of-Healthcare-Things (IoHT) networks because it allows clients to train collaboratively without centralizing sensitive data. However, client model updates can still leak information about local training records. Differential privacy (DP) can reduce this risk, but fixed privacy settings may be poorly suited for heterogeneous IoHT clients with non-IID data.**

In this work, we evaluate fixed and adaptive client-side privacy mechanisms for federated IoHT intrusion detection. We simulate heterogeneous clients using Dirichlet-based partitioning on the ECU-IoHT and WUSTL-EHMS 2020 datasets, and compare no-DP training, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping. We evaluate detection utility using precision, recall, F1-score, ROC-AUC, and PR-AUC, and assess privacy leakage using a white-box membership inference attack.

Our results show that strict no-identifier preprocessing reduces membership inference performance close to chance across privacy mechanisms. Utility-preserving DP settings require very small noise multipliers, corresponding to weak approximate per-update privacy scales, while stronger target epsilon settings require much larger noise and cause utility collapse. Overall, adaptive mechanisms provide utility comparable to fixed-noise DP in our evaluated settings, suggesting that adaptive privacy has limited room to improve when empirical leakage is already low and stronger noise is not utility-tolerable.

***Index Terms*—Federated learning, differential privacy, IoHT, intrusion detection, membership inference, non-IID data, privacy–utility tradeoff**

I. Introduction

Internet-of-Healthcare-Things (IoHT) devices are increasingly used to monitor patients, transmit medical data, and support connected healthcare services. The data generated by these systems can improve machine learning-based intrusion detection, but centralizing sensitive medical and network data raises privacy concerns. Federated learning (FL) reduces this risk by allowing clients to train locally and share model updates rather than raw data. However, model updates can still leak information about local training samples.

Differential privacy (DP) is commonly used to reduce such leakage by perturbing client updates before aggregation. Many DP-based FL methods apply a fixed noise level to all clients, but this is limiting for heterogeneous IoHT clients with different data volumes, class balances, and traffic distributions. In intrusion detection, where malicious samples may be rare and unevenly distributed, a single fixed privacy setting can produce an unfavorable privacy–utility tradeoff.

Adaptive privacy mechanisms address this limitation by adjusting perturbation across clients or training rounds. Prior work has shown the promise of adaptive local DP in FL, but often on general image-classification benchmarks rather than heterogeneous IoHT intrusion-detection datasets. Thus, the practical benefit of adaptive privacy for federated IoHT intrusion detection remains under-characterized.

In this work, we evaluate fixed and adaptive client-side privacy mechanisms for federated IoHT intrusion detection. We use Dirichlet-based partitioning to simulate non-IID clients on ECU-IoHT and WUSTL-EHMS 2020, compare fixed and adaptive privacy mechanisms, and evaluate both IDS utility and white-box membership leakage.

Our main contributions are as follows:

- We evaluate fixed and adaptive client-side privacy mechanisms under heterogeneous, non-IID federated IoHT clients.
- We compare privacy–utility behavior on ECU-IoHT and WUSTL-EHMS 2020 using IDS metrics and white-box membership inference.
- We find that adaptive privacy mechanisms provide utility comparable to fixed-noise DP in our current settings, while membership inference results are strongly influenced by preprocessing choices such as identifier removal.

The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the methodology. Section IV describes the experiments and results. Section V concludes the paper.

II. Related Work

Federated learning has been widely studied for distributed machine learning when raw data cannot be centralized. In healthcare and IoHT environments, FL is attractive because medical and network traffic data may contain sensitive information. However, FL alone does not guarantee privacy, since model updates can still reveal information about local training records.

Differential privacy is commonly used to mitigate this risk by clipping model updates and adding calibrated noise before aggregation. Fixed-noise DP methods are simple, but may be poorly calibrated when clients differ in data size, label distribution, and update behavior. Adaptive privacy mechanisms attempt to improve this tradeoff by changing clipping thresholds or noise levels across clients or training rounds.

Membership inference attacks provide a practical way to evaluate empirical privacy leakage. In these attacks, an adversary attempts to determine whether a specific record was included in the model's training data. White-box attacks are especially relevant in federated settings because the adversary may observe model parameters, gradients, losses, or other training-related signals.

III. Methodology

A. System Model

We consider a federated IoHT intrusion-detection setting with K heterogeneous clients. Each client represents a local healthcare environment, such as a glucose-monitoring system, blood-pressure monitoring system, or patient-monitoring system. Client k holds a private local dataset D_k, which may differ from other clients in size, class balance, and traffic distribution. Raw IoHT traffic and training records remain local throughout training.

A central federated server coordinates training but does not receive raw data. In each communication round, clients train local IDS models, compute model updates, apply the assigned privacy mechanism, and send only the resulting updates to the server. The server aggregates these updates to produce a global IDS model, which is then redistributed to the clients.

B. Federated Learning Formulation

Let θ^t denote the global model parameters at round t. The server broadcasts θ^t to the participating clients. Each client trains locally and obtains updated parameters θ_k^t. The client update is

$$\Delta_k^t = \theta_k^t - \theta^t.$$

The server aggregates client updates using FedAvg:

$$\theta^{t+1} = \theta^t + \sum_{k=1}^{K} w_k \widetilde{\Delta}_k^t,$$

where w_k is the aggregation weight for client k, and $\widetilde{\Delta}_k^t$ is the update received by the server. In the no-DP setting, $\widetilde{\Delta}_k^t = \Delta_k^t$. In DP-based settings, the update is clipped and noised before upload.

C. Client-Side Privacy Mechanisms

For DP-based methods, each client update is first clipped:

$$\bar{\Delta}_k^t = \Delta_k^t \cdot \min\left(1, \frac{C_t}{\|\Delta_k^t\|_2}\right),$$

where C_t is the clipping threshold. Gaussian noise is then added:

$$\widetilde{\Delta}_k^t = \bar{\Delta}_k^t + \mathcal{N}\left(0, \sigma_{k,t}^2 C_t^2 I\right).$$

This follows the standard Gaussian mechanism: for $\epsilon \in (0,1)$, if $c^2 > 2\ln(1.25/\delta)$, adding Gaussian noise with standard deviation at least $c\Delta_2(f)/\epsilon$ gives (ϵ, δ)-DP. In our implementation, clipping bounds the update scale through C_t, while the Gaussian noise is controlled by the multiplier $\sigma_{k,t}$. Larger $\sigma_{k,t}$ therefore corresponds to a smaller approximate ϵ, but also stronger perturbation of the client update. We use this relationship as a per-update privacy scale, not as a composed privacy accountant across FL rounds.

We compare three DP variants. In fixed-noise DP, all clients use the same noise multiplier, $\sigma_{k,t} = \sigma$. In smoothed adaptive noise scaling, the noise multiplier changes using a smoothed update-norm estimate:

$$s_{k,t} = \lambda s_{k,t-1} + (1-\lambda)\left\|\Delta_k^t\right\|_2,$$

$$\sigma_{k,t} = \mathrm{clip}\left(\sigma_0\left(1 + \beta \log\left(1 + \frac{s_{k,t}}{C_t}\right)\right), \sigma_{\min}, \sigma_{\max}\right).$$

In adaptive clipping, the noise multiplier remains fixed while the clipping threshold is adjusted during training.

IV. Experimental Setup and Results

A. Datasets and Preprocessing

We evaluate on ECU-IoHT and WUSTL-EHMS 2020. ECU-IoHT serves as a comparatively separable benchmark in our experiments, while WUSTL-EHMS 2020 provides a more sensitive test of privacy perturbation and class imbalance.

For the main evaluation, we apply strict no-identifier preprocessing by removing identifier-like fields such as IDs, MAC-like features, and other direct or quasi-identifiers. This is important because identifier-like features can create leakage unrelated to the federated privacy mechanism itself. We also evaluate an identifier-inclusive WUSTL setting as a leakage-positive control to verify that the membership inference attack detects leakage when leaky features are present.

B. Federated Client Partitioning

Because both datasets are centralized, we simulate federated clients using label-based Dirichlet partitioning. For each class, samples are distributed across K clients according to

$$p \sim \mathrm{Dirichlet}(\alpha).$$

The concentration parameter α controls the degree of non-IID behavior. Smaller values create stronger client heterogeneity. We use $K = 8$ clients and $\alpha = 0.3$ to create uneven client sizes and class balances, reflecting the heterogeneous-client setting targeted by this study. Fig. 2 shows the resulting client class composition.

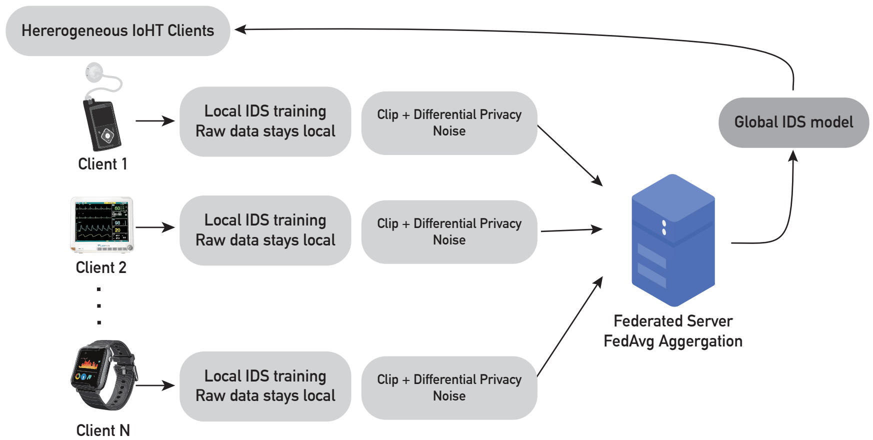

Fig. 1. System model for federated IoHT intrusion detection with heterogeneous clients. Raw data remain local to each IoHT environment, while privatized model updates are sent to the federated server for aggregation.

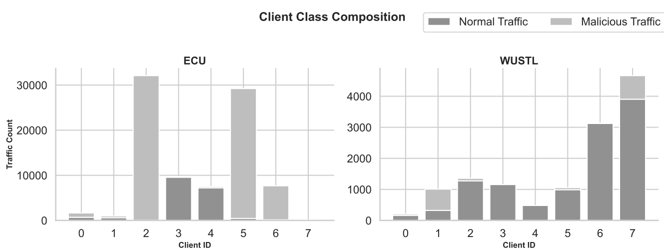

Fig. 2. Dirichlet-based federated client split for ECU-IoHT and WUSTL-EHMS 2020 using $K = 8$ clients and $\alpha = 0.3$. The split creates heterogeneous client sizes and class compositions.

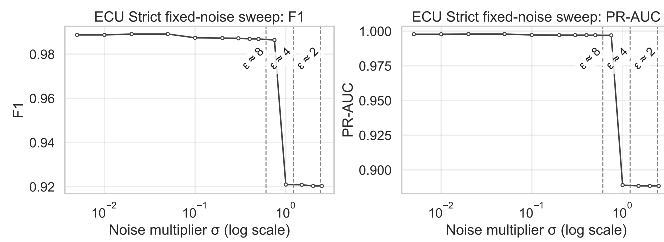

Fig. 3. Fixed-noise utility sweep for ECU-IoHT under strict no-identifier preprocessing. F1-score and PR-AUC remain high across the tested noise multipliers, suggesting that ECU-IoHT is comparatively separable in the current setting.

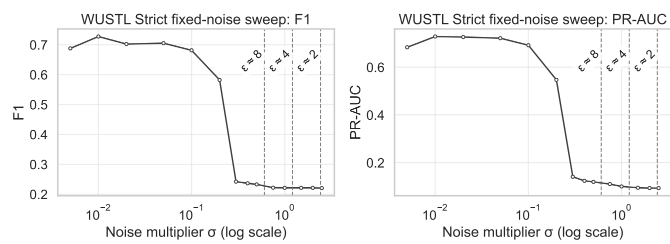

Fig. 4. Fixed-noise utility sweep for WUSTL-EHMS 2020 under strict no-identifier preprocessing. F1-score and PR-AUC are lower and more sensitive to the noise multiplier, showing a clearer utility cost from privacy perturbation.

C. Training Conditions and Metrics

We compare four training settings: no-DP training, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping. All settings use the same client splits, model architecture, and FL pipeline so that differences are attributable to the privacy mechanism.

We evaluate IDS utility using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Because IoHT intrusion-detection data are imbalanced, we emphasize F1-score, ROC-AUC, and PR-AUC rather than accuracy alone. We evaluate privacy leakage using white-box membership inference ROC-AUC, where values near 0.5 indicate chance-level attack performance.

D. Fixed-Noise Utility Sweep

To understand how each dataset responds to increasing privacy noise, we perform a fixed-noise sweep under strict no-identifier preprocessing. This sweep varies the Gaussian noise multiplier and measures F1-score and PR-AUC.

Fig. 3 shows that ECU-IoHT remains near the utility ceiling across the tested noise range. Both F1-score and PR-AUC remain high, indicating that the dataset is relatively separable under the current preprocessing and model configuration. This limits the room for any privacy mechanism, fixed or adaptive, to improve measured utility.

Fig. 4 shows a different pattern for WUSTL-EHMS 2020. Utility is lower overall and changes more substantially as the noise multiplier increases. This indicates that WUSTL-EHMS 2020 is more sensitive to privacy perturbation and therefore exposes the utility cost of increasing DP noise more clearly.

E. Membership Inference Results

Fig. 5 shows the white-box membership inference ROC-AUC under strict no-identifier preprocessing. For both ECU-IoHT and WUSTL-EHMS 2020, attack performance remains close to the chance line across no-DP, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping. This

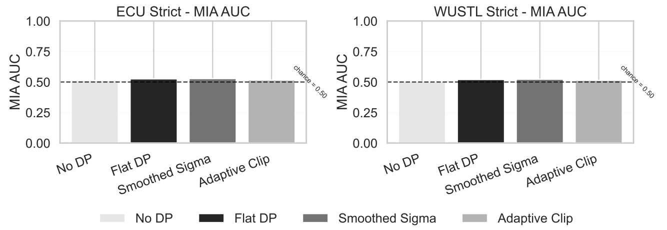

Fig. 5. White-box membership inference ROC-AUC under strict no-identifier preprocessing. Attack performance remains close to chance for both ECU-IoHT and WUSTL-EHMS 2020 across no-DP, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping.

TABLE I
NOISE MULTIPLIER AND APPROXIMATE GAUSSIAN-MECHANISM PRIVACY SCALE. EPSILON VALUES ARE APPROXIMATE PER-UPDATE ESTIMATES AT $\delta = 10^{-5}$ AND ARE NOT COMPOSED ACROSS FL ROUNDS.

Method	**Mean σ**	**Approx. ϵ**
Fixed noise	0.010	484.5
Smoothed sigma	0.011	441–458
Adaptive clip	0.010	484.5
Target $\epsilon = 2$–8	≈ 0.43–1.65	2–8

indicates that, after identifier removal, there is little measurable membership leakage under the evaluated attack.

This result is central to interpreting the DP experiments. Since no-DP training is already near chance under strict preprocessing, DP has limited room to further reduce empirical membership inference risk. Therefore, changes in the privacy mechanism are mainly reflected in utility behavior rather than a large change in measured MIA AUC.

F. Approximate Privacy Scale

Table I summarizes the approximate Gaussian-mechanism privacy scale used to interpret the tested noise multipliers. The utility-preserving settings use very small noise multipliers around $\sigma \approx 0.010$, which correspond to very large approximate per-update ϵ values. These settings preserve IDS utility but provide a weak formal privacy scale under the Gaussian mechanism estimate.

We also tested whether more conventional target ϵ values could be used by increasing the noise multiplier. Targeting $\epsilon = 2$–8 required much larger noise multipliers, approximately 0.43–1.65, and these settings caused utility to collapse. At the same time, membership inference remained near chance because strict no-identifier preprocessing had already removed most measurable leakage. Thus, simply increasing privacy noise did not reveal a useful adaptive-DP regime: the stronger settings damaged utility, while the empirical MIA signal remained low.

G. Fixed and Adaptive Privacy Comparison

Across the evaluated settings, adaptive privacy mechanisms provide utility comparable to fixed-noise DP. ECU-IoHT remains near ceiling across mechanisms, while WUSTL-EHMS 2020 is more sensitive to DP perturbations. The fixed-noise sweeps in Fig. 3 and Fig. 4 show that increasing noise is not equally tolerable across datasets.

The MIA results in Fig. 5 explain why adaptive mechanisms do not show a large empirical privacy advantage in the current setting. After strict preprocessing, measured membership leakage is already near chance, so changing the privacy mechanism does not substantially change attack AUC. The epsilon sensitivity in Table I further shows that increasing noise enough to target stronger privacy scales is not utility-tolerable. Together, these results identify a specific limitation: adaptive DP had little room to improve privacy on these cleaned datasets because empirical leakage was already low, while stronger perturbation destroyed utility.

V. CONCLUSION

This paper evaluated fixed and adaptive client-side privacy mechanisms for federated IoHT intrusion detection under heterogeneous non-IID clients. We compared no-DP training, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping on ECU-IoHT and WUSTL-EHMS 2020, and evaluated privacy leakage using a white-box membership inference attack.

Our results show that strict no-identifier preprocessing reduces membership inference performance close to chance across all evaluated privacy mechanisms. In terms of utility, ECU-IoHT remains near ceiling, while WUSTL-EHMS 2020 is more sensitive to privacy noise. Utility-preserving DP settings require very small noise multipliers, while target $\epsilon = 2$–8 settings require much larger noise and collapse utility. Therefore, adaptive mechanisms provide utility comparable to fixed-noise DP in this study not simply because the datasets are insufficiently challenging, but because the evaluated regime leaves little empirical membership leakage to reduce and little utility tolerance for stronger perturbation.

Future work should focus on settings where adaptive behavior can be evaluated more directly, such as drift-based adaptive clipping, client-specific update diagnostics, and benchmarks or splits with measurable leakage after preprocessing but enough utility tolerance for meaningful DP noise.

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